SUMIT SARASWAT

Machine Learning · AI Research · Reinforcement Learning · Computational Biology · Algorithmic Trading

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ACHIEVEMENTS & AWARDS

🏆 **Meta × PyTorch OpenEnv Hackathon — Grand Finalist** (Top ~0.1% of 52,000+ registrants)

- Built SynthAudit.Env — GRPO-trained multi-agent clinical AI oversight environment; +283% improvement over Qwen2.5-3B base model, 4× more correct error flags from zero supervised data
- Outperformed Meta Llama 3.1 405B (strongest open-source frontier model) across all difficulty tiers — Avg 0.66 vs 0.50, Precision 45% vs 26%
- Presented to Meta & Hugging Face AI engineers at on-campus Grand Finale, Bangalore, April 2026. Competing for $30,000 prize pool + direct interview opportunity

🥇 **INSPIRE MANAK Award — Government of India, National Top 3**

- Highest civilian recognition for student innovation; selected from 700,000+ national entries by DST, Government of India

🔬 **INSPIRE Summer Research Fellowship (SRFP) — DST, Government of India**

- Competitive national fellowship awarded to outstanding science students for independent research

EDUCATION

B.Tech Computer Science Engineering · GLA University, Mathura *Aug 2025 – May 2029*

- First year B.Tech CSE student with strong focus on ML research, AI systems, and quantitative applications

ML & AI RESEARCH PROJECTS

SynthAudit.Env — Multi-Agent RL · GRPO Training · Meta × PyTorch OpenEnv Grand Finale *2026*

- Built multi-agent adversarial RL environment — one AI generates deceptive medical errors, another learns to catch them via 8 investigation tools
- Designed 4 adversarial error layers requiring multi-hop reasoning, Theory-of-Mind scoring, dense shaped rewards (F-β, β=1.5), and adaptive curriculum with procedural EHR generation (40-80 patients/episode)
- GRPO-trained Qwen2.5-3B-Instruct (200 steps, free Colab T4, $0 compute) — +283% improvement over base, peak reward 0.506, model learned full ReAct audit chains from zero supervised data
- Outperformed Meta Llama 3.1 405B across Easy/Medium/Hard tiers. Passed openenv validate. Deployed on Hugging Face Spaces

  github.com/sumitsaraswat362/SynthAudit.Env

Clinical Trial Auditor RL Environment — Reinforcement Learning · Meta × PyTorch OpenEnv Round 1*2025*

- Custom OpenAI Gym-compatible RL environment for auditing clinical trial protocol compliance — evolved into SynthAudit.Env for Grand Finale
- ReAct agent outperformed Meta Llama 3.1 405B (Avg 0.66 vs 0.50, Precision 45% vs 26%) across all difficulty tiers
- Passed Round 1 validation out of 52,000+ submissions. Built with PyTorch, Meta OpenEnv, Hugging Face Transformers

  github.com/sumitsaraswat362/clinical-trial-auditor

Equity-Aware Survival Analysis — Computational Oncology · Manuscript in Preparation *2024 – Present*

- Independent research auditing breast cancer survival models on 170,000+ SEER registry patients (50,355 analyzed)
- **0.746** vs Cox baseline 0.733 under strict temporal external validation (2004–2015 train, 2016–2018 test)
- Identified clinically catastrophic failure modes — false-negative rate 34× higher in predicted low-risk group; demographic subgroup fairness audit across racial groups

  github.com/sumitsaraswat362/equity-aware-survival-analysis

BTC Momentum Trading Strategy — Algorithmic Trading · Systematic Backtesting *2025*

- MA20/MA50 crossover with volatility-adjusted position sizing — Sharpe Ratio 3.26, Total Return 72.19%, Max Drawdown −41.62%, Win Rate 52% (2022–2026)
- Walk-forward validation (70/30 split), zero lookahead bias; outperformed Buy & Hold through 2022 bear market
  github.com/sumitsaraswat362/btc-momentum-strategy

TECHNICAL SKILLS

Languages	Python, JavaScript
ML Frameworks	PyTorch, scikit-learn, OpenAI Gym / Gymnasium, Meta OpenEnv, Hugging Face Transformers, TRL, Unsloth, SHAP
ML & AI	Reinforcement Learning (GRPO, custom Gym environments, reward shaping, ReAct agents), LLM fine-tuning, LLM benchmarking & evaluation, LangGraph, RAG pipelines, Random Survival Forest, Cox Proportional Hazards, survival analysis, subgroup fairness auditing, SHAP interpretability
Data & Quant	NumPy, Pandas, matplotlib, yfinance, statistical modeling, temporal validation, Sharpe ratio, C-index, Brier score, ROC-AUC, precision/recall
Deployment	Hugging Face Spaces, Docker, FastAPI, Git/GitHub, reproducible ML pipelines

COMPETITIVE & RESEARCH ACTIVITIES

Meta × PyTorch OpenEnv Hackathon — Grand Finalist (Top ~0.1% of 52,000+ devs, Solo Entry) *2026*